

Signals Protocol Research Notes

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Abstract

Concise summary of the Signals Protocol research direction. Replace this paragraph with a crisp overview of the problem, approach, and expected outcomes.

1 Introduction

Signals is an on-chain protocol that converts latent community sentiment into a real-time, ranked idea board of DAO initiatives. Inspired by conviction voting¹, commitment voting², and veToken systems such as Curve, Aragon, and Velodrome³, it lets members lock their existing governance tokens for a period of time to signal their support for what they think is most important. This token-time locking, coupled with configurable decay curves and acceptance thresholds, equips decentralized organizations with a transparent, mathematically grounded mechanism for continuous sensemaking.

We expect Signals to become an on-chain instrument for prioritizing structural changes and capital allocation while creating new utility for governance tokens.

1.1 Goals

- Deliver correctness and liveness guarantees under realistic network partitions.
- Keep the trust surface minimal and explicit.
- Make operational costs predictable for deployments of varying scale.

1.2 Non-goals

List out-of-scope items to avoid scope creep and clarify assumptions.

2 Motivation

Token-based voting captures binary preferences on formal proposals but does little to surface intensity. After a decade of on-chain experimentation, DAOs still struggle to translate intent into timely, accountable action. Common pain points include:

- Endless discussion threads with unclear next steps.

¹[1, 6].

²[3, 7].

³[5, 4, 2, 8].

- Committees that drift or concentrate power.
- Fragmented knowledge bases that deter newcomers.
- Social pressure that favors incumbents over fresh ideas.

By leveraging preference intensity through time-lock commitments, SIGNALS constructs a dynamic idea board where the most strongly supported initiatives rise naturally, keeping attention on what matters and when.

3 Related Systems

Conviction Voting Holders stake without fixed locks; conviction builds the longer tokens stay committed and can be re-allocated at any time (which resets build-up). Proposals auto-execute once time-weighted conviction crosses an adaptive threshold tied to requested resources. It channels persistent conviction into binding, resource-allocating decisions.

Commitment Voting Voters lock tokens for a period to determine voting power. If their choice wins, tokens remain locked as committed; losing side unlocks immediately. Victory carries opportunity cost, so intensity matters. See [3, 7].

veToken Systems veCRV, Aragon OSx gauges, and Velodrome require duration-based locks to earn voting power. Lock length boosts influence across all votes but does not capture intensity on a specific initiative. See [5, 4, 8, 2].

4 Mechanism Design

4.1 Initiatives, not Proposals

SIGNALS works with *Initiatives*—ideas or actions seeking support. Initiatives live inside Boards (configuration sets), and DAOs can run multiple Boards with different parameters. Board configs are on-chain, so DAOs can govern them directly.

4.2 Token-Locking

Participants lock existing tokens for a chosen duration to express commitment; locking against an Initiative is not supported. Each lockup mints a transferable NFT bond redeemable when the lock expires, enabling secondary markets or other reputation measures. Boards typically use governance tokens but can target any ERC-20.

4.3 Calculating Support

Time is measured in Epochs (Board-defined). Longer locks provide more weight:

$$S = T \times D^k \tag{1}$$

where S is support, T is tokens locked, D is duration in Epochs, and k is an exponential bonus favoring longer commitments. Tuning k prevents short, large locks from dominating and lets smaller holders compete by committing longer.

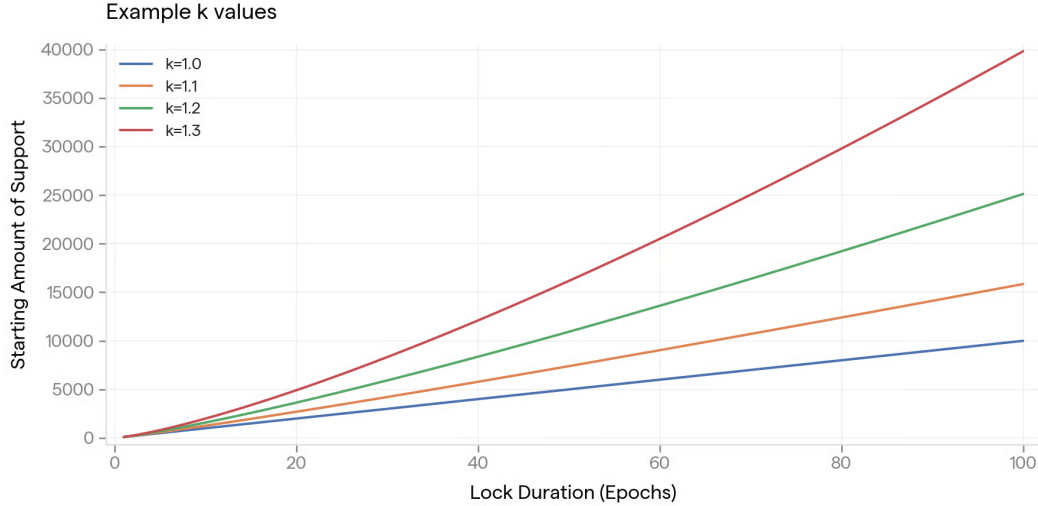


Figure 1: Adjusting k changes how much additional weight longer lockups receive.

4.4 Decay

Support decays over time to keep rankings current. Each Board specifies a decay curve (linear, exponential, or custom). Current support for Initiative i at time t sums decayed contributions from its locks ℓ :

$$\text{CurrentSupport}_i(t) = \sum_{\ell \in \text{Locks}_i} \text{Weight}(\ell, t). \quad (2)$$

Decay ensures abandoned initiatives fade, early interest provides a head start (not permanence), and manipulation via stale locks is limited.

4.5 Actioning Initiatives

Boards define an acceptance threshold. When current support exceeds it, an Initiative can be actioned: the Initiative closes and all supporting tokens unlock early. DAOs decide whether actioning is automatic or requires a trigger (committee, vote, or other mechanism). The early unlock incentivizes backing only initiatives expected to succeed.

4.6 Examples of Support and Thresholds

4.7 Endorsements (Optional)

If a DAO uses delegated voting, delegates may not own the tokens they wield. Boards can optionally require binary *Endorsements* tied to delegated voting power in addition to lockups. Caps on endorsements per Initiative encourage delegates to act early and visibly.

5 Paid Incentives in Signals

5.1 Motivation

DAOs often face low participation: most holders do not see sufficient upside to engage, and reputational risk pushes signaling to the last minute. Some DAOs spend \$1M+ per year on delegate

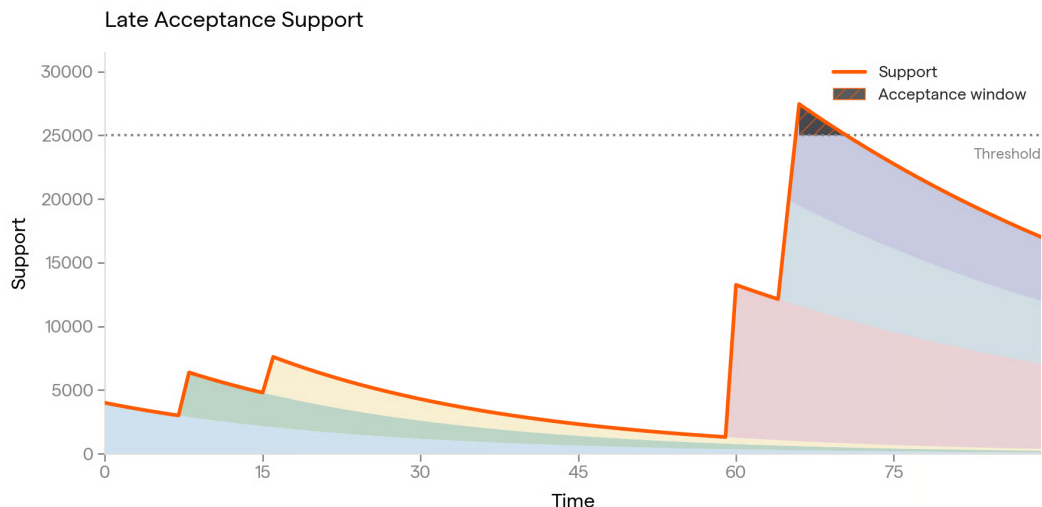


Figure 2: Early support is insufficient; renewed interest later can re-open the path to action.

programs. Signals inherits similar participation risks, so we add configurable incentives that reward early, accurate support and penalize weak or misaligned signaling.

5.2 Mechanism 1: General Participation Incentives

A Board can be funded (e.g., from treasury) to share rewards with supporters of any Initiative that gets actioned. Reward share can be time-weighted so early supporters earn more:

- Improves effort-reward ratio to motivate broader participation.
- Competes for early signaling instead of last-minute pile-on.
- Supporting an unpopular Initiative risks missing rewards elsewhere, encouraging intentional choices.

5.3 Mechanism 2: Sponsored Initiative Incentives

Anyone can add rewards to a specific Initiative. If it expires, rewards revert to the sponsor; if actioned, rewards are split among supporters, treasury, and protocol according to a configured ratio (e.g., 80/15/5). Benefits:

- Sponsors signal conviction by staking funds on success.
- External partners can prove commitment to a DAO initiative up front.
- Popular initiatives can crowd-source additional rewards to amplify sentiment.

5.4 Impact

Rewards amplify Signals core design: only positive support, opportunity cost via locks, and time-weighted influence that lets smaller holders compete. Incentives raise the cost of frivolous signaling and make early, accurate support economically meaningful, improving resilience against brigading or apathy-driven noise.

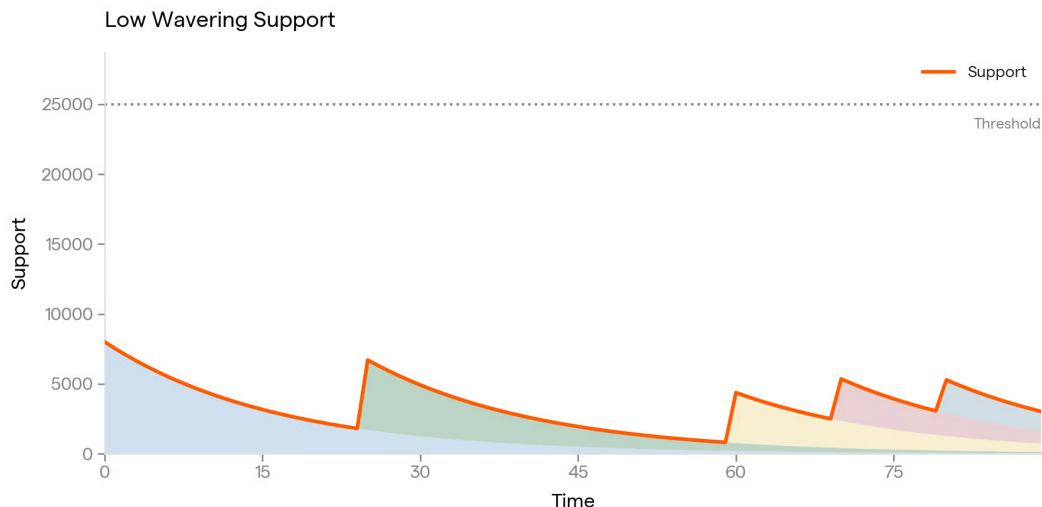


Figure 3: Insufficient, wavering support never crosses the threshold despite cumulative effort.

6 What's Next

The Signals proof of concept was built during the Arbitrum Collabtech Hackathon (1st place) and continued in the Uniswap Hooks Incubator (prizes from Arbitrum and the Uniswap Foundation). Contracts and dashboard are open source.⁴

We aim to formalize and validate the protocol on:

- Capturing voter preference intensity and opportunity cost as sufficient risk.
- Locking as sybil resistance and its limits.
- Gameable vs. productive patterns for endorsements and reputation.
- Inclusivity: how smaller voting blocks benefit from time-weighted designs.

If interested in collaborating on a whitepaper, please reach out.

References

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⁴<https://github.com/OxLighthouse/signals>

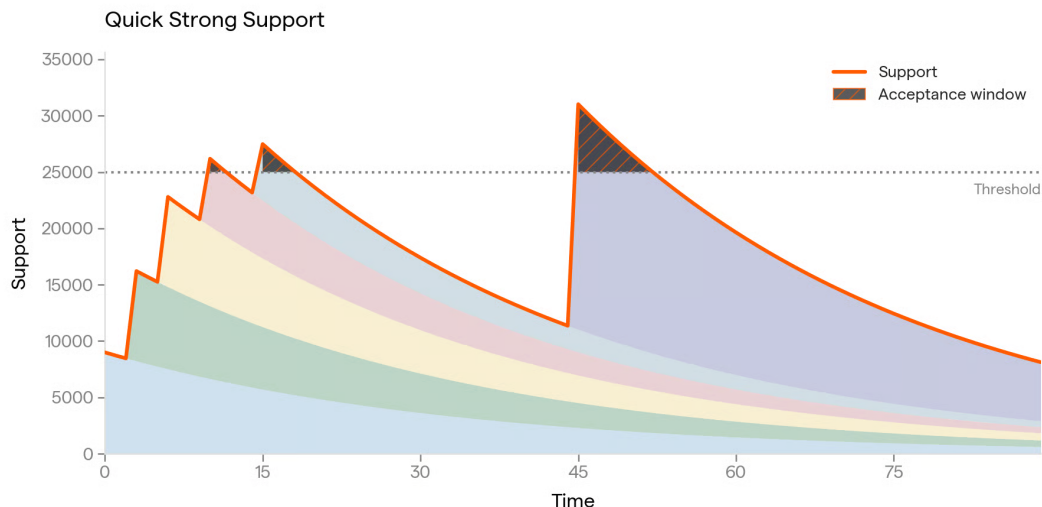


Figure 4: Late momentum pushes an Initiative over the threshold after an initial lull.

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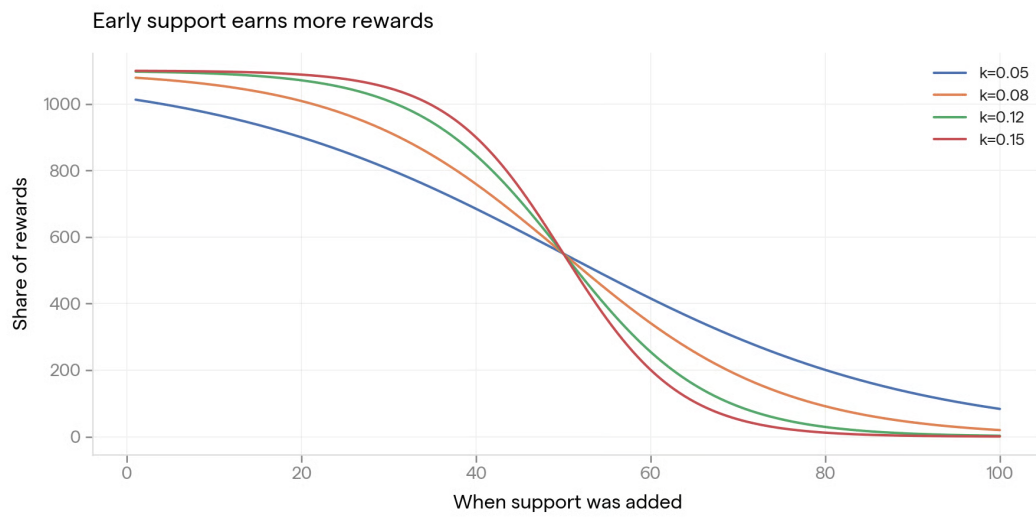


Figure 5: Example reward distribution emphasizing early supporters and shared upside.